SELF-X

SDN control with state embedding memory

Measured simulation with live Gemini and free OpenRouter decisions

20 October 2026

Project purpose

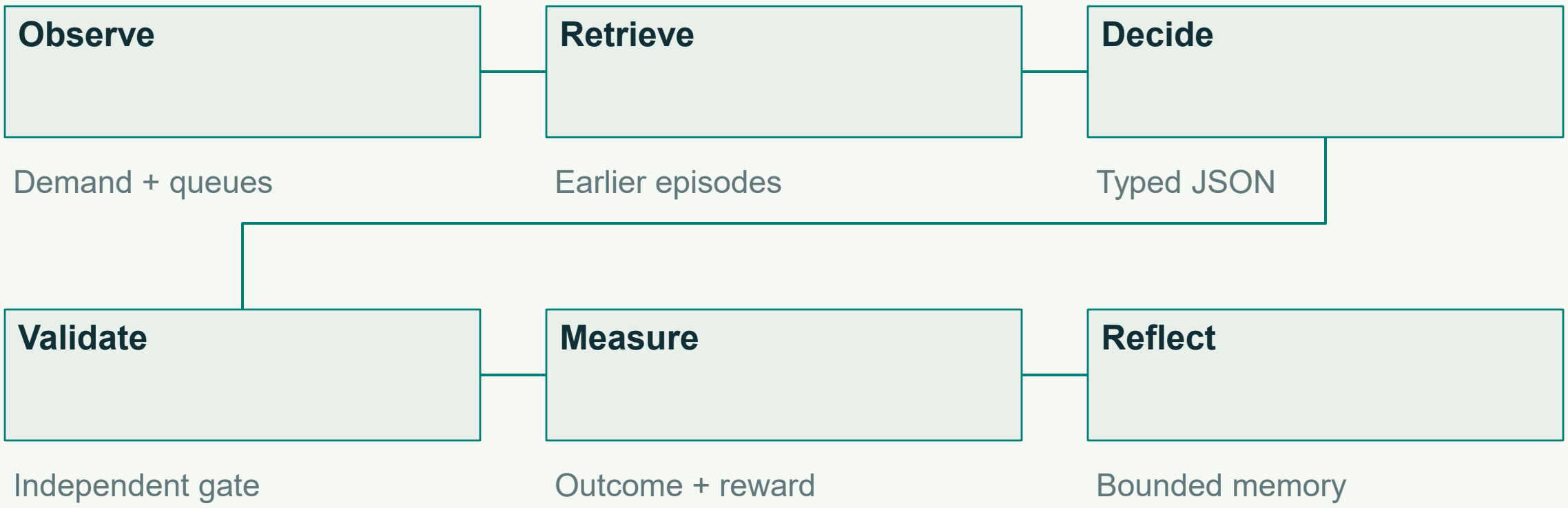
Can a controller reuse its own experience to manage repeated congestion?

Observe traffic and choose one bounded flow intervention.

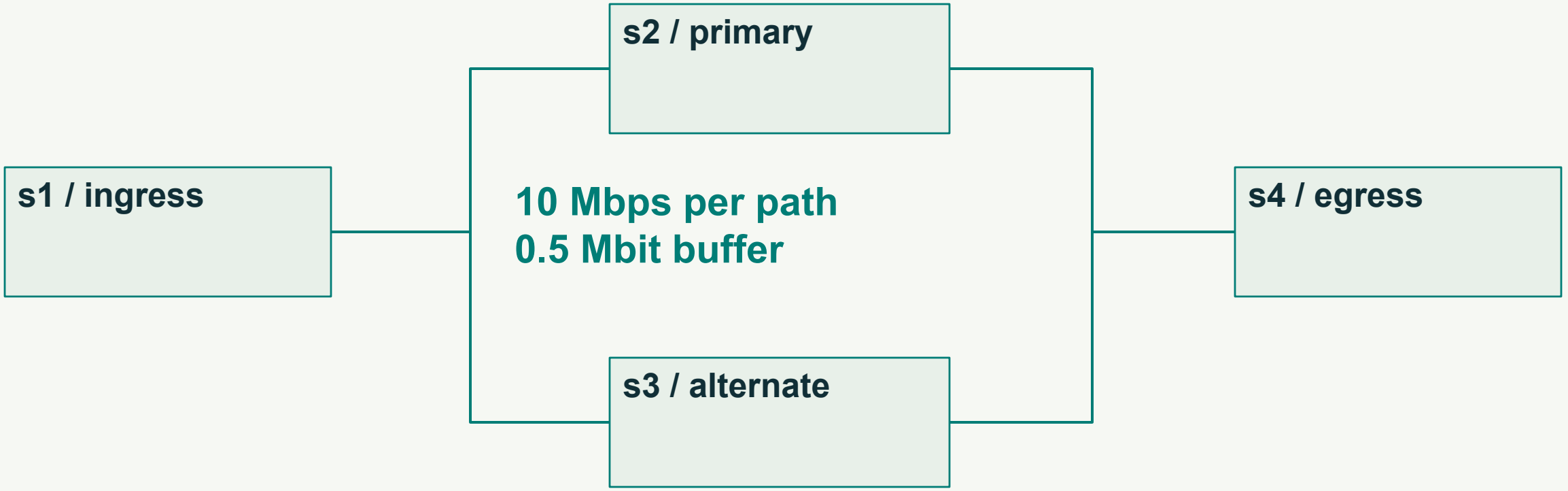
Retrieve earlier states, outcomes and reflections before deciding.

Use the local website to compare results and inspect every action.

Control architecture



Two-path fluid topology



8 ms propagation component; five-second windows; one elephant flow plus path-specific background traffic

A logical network model; no packet testbed is claimed

State embedding and episodic memory

Traffic demand + queue occupancy

Normalize features → unit vector → cosine similarity

Retrieve four earlier episodes from a memory of at most 30.

Store state, installed action, reward and a concise reflection.

Weights remain fixed. The embedding is a hand-designed numeric representation.

Actions and deterministic safety

Action	Required checks
Reroute	Exact flow and switch; alternate load ≤ 90% capacity
Rate limit	At most 20% reduction; minimum 512 kbps
No-op	Safe fallback for invalid actions or provider failure

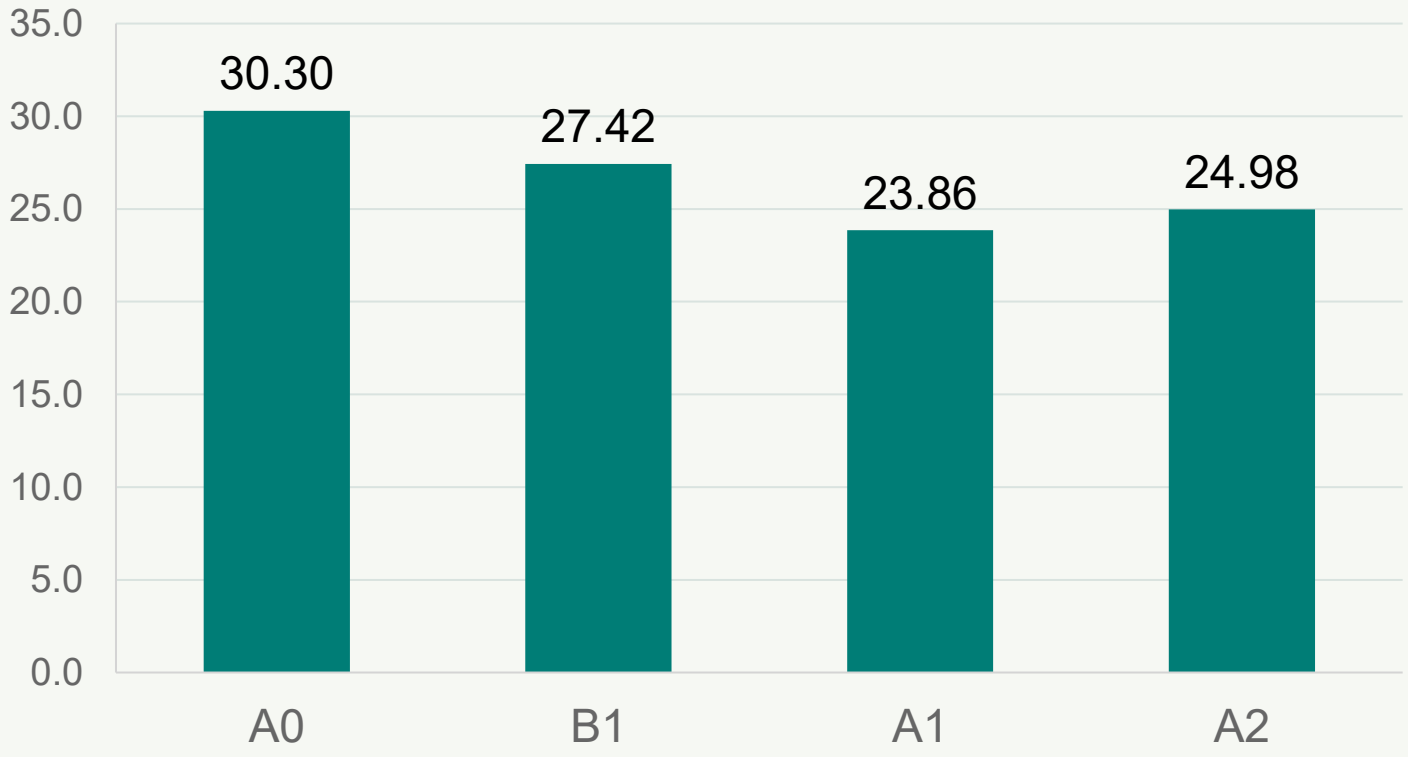
Typed output is necessary. A separate gate decides whether the action can be applied.

Experimental design

Experiment	Seeds	Windows each	A1 and A2 backend
Portable study	5	120	Deterministic policies
Paced Gemini	3	24	Real Gemini 3.5 Flash-Lite
Free-router diagnostic	1	12	Real free model or visible fallback

A0: static route B1: feasible least-loaded route
A1: no episode memory A2: similar-state memory

Offline latency results



Memory did not improve the offline controller

A1: 23.86 ms
A2: 24.98 ms

A2 minus A1:
+1.13 ms

Five seeds; 600 windows per controller; modeled delay in ms

Goodput and total loss

Controller	Goodput Mbps	Mean total loss %	Safety vetoes
A0	13.326	9.966	0
B1	13.399	9.416	0
A1	13.308	10.013	0
A2	13.233	10.691	62

A2 traded lower delay than A0 for lower goodput and more loss.
Latency alone is insufficient evidence of a better controller.

Mean window loss includes deliberate meter drops

Interpreting reflection and memory

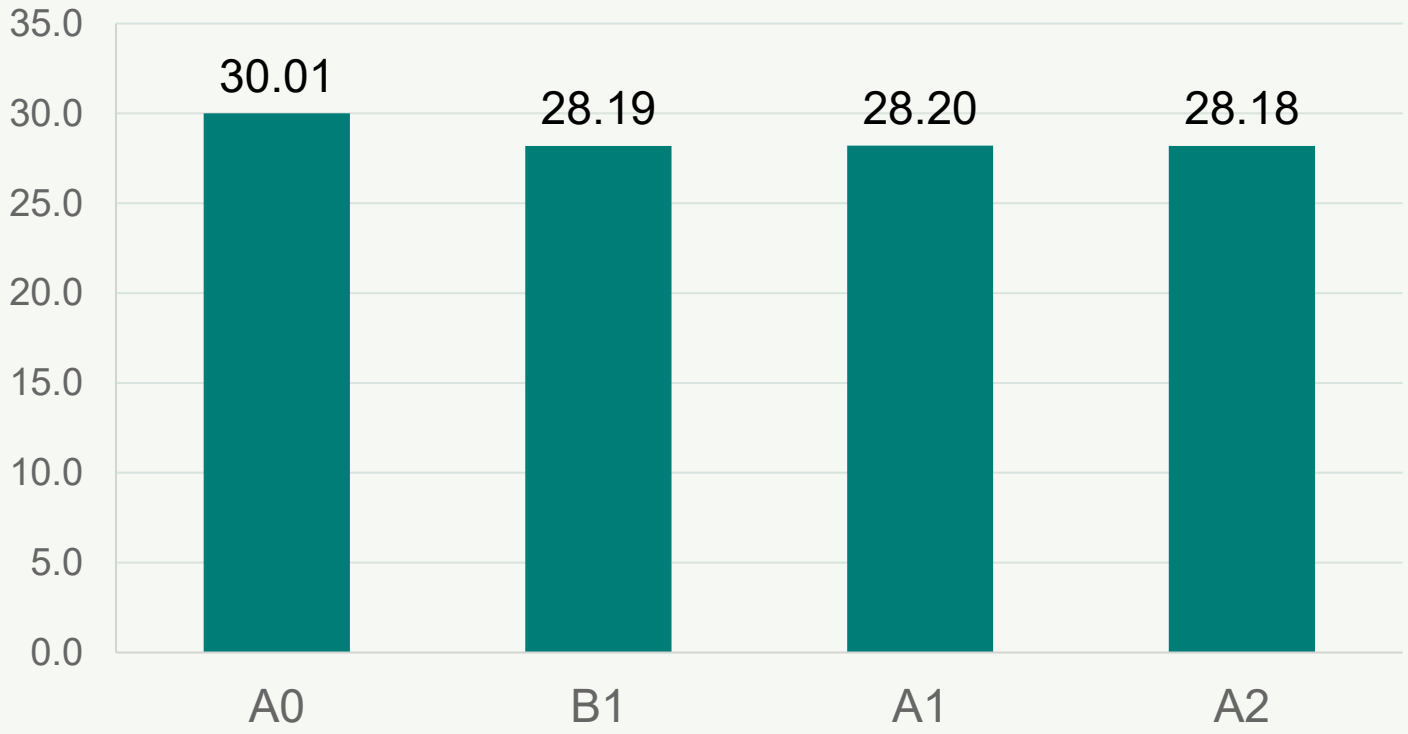
62 vetoes in the offline memory condition

A similar state can require a different feasible action.

Reward pruning can discard informative failed episodes.

A stored explanation supports adaptation; it does not guarantee improvement.

Gemini integration results



288 / 288

successful decision and reflection API calls

A1 decisions: 72/72
A2 decisions: 72/72

Vetoes: 48 → 22

A2 latency differs from A1 by -0.017 ms; goodput and loss are equal.

Three seeds × 24 windows per controller; network results are simulated

Free model availability

9 / 17

models returned valid decisions

8 checks were unavailable or unsuitable in this run.

Free-router diagnostic:
A1 API decisions 3/12
A2 API decisions 0/12

Availability and JSON compatibility are separate from control quality.

One structured healthy-network check per discovered zero-price model ID

Related work

Work	Relevance	Comparison boundary
Reflexion (2023)	Verbal feedback and episode memory	Different tasks; no SDN score transfer
RouteNet (2019)	Topology-aware network KPI modeling	15.4% worst-case MRE is prediction error
Local A0 and B1	Static and feasible greedy baselines	Measured on the same demand

Published measurements retain their original task and units.
They are not substituted for local experimental results.

Engineering corrections and validation

Outputs and experiment paths stay inside selfx.

Metered OpenFlow rules preserve observed forwarding.

Rerouting requires a validated alternative egress.

Tests cover conservation, reproducibility, provider fallback and dashboard access.

Limitations and next experiments

The fluid model is not packet capture or a physical-network measurement.

The hand-designed embedding has no formal learning or stability guarantee.

Paced live API calls do not meet a five-second real-time deadline.

Next: endpoint-instrumented Linux tests, fixed models, more workloads and topologies.

Expanded benchmark coverage

57,600

measured control windows

Traffic scenarios	Paired seeds	Windows per condition
6	10	120 per seed
8 controllers	600 simulated seconds	per scenario and seed

Includes original A2, guarded H1 and A3, plus a myopic model reference.

Offline decisions; all network values are simulated

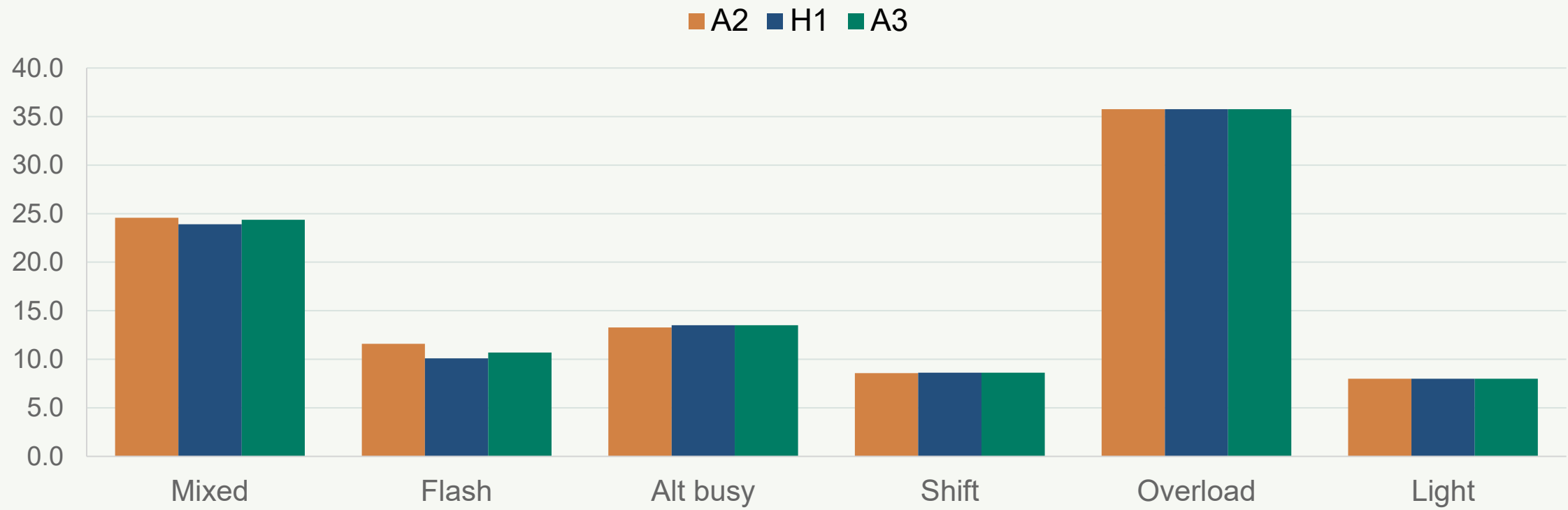
Scenario delay benchmark

Scenario	A0	B1	B2	A1	A2	H1	A3	O1
Mixed	30.45	27.38	28.79	23.73	24.57	23.90	24.37	23.77
Flash	26.37	10.80	18.76	10.04	11.60	10.09	10.70	10.04
Alt busy	21.82	21.82	21.82	13.29	13.28	13.51	13.51	13.63
Shift	25.58	15.21	19.93	8.57	8.57	8.62	8.62	8.98
Overload	35.77	35.77	35.77	35.77	35.77	35.77	35.77	35.77
Light	8.00	8.00	8.00	8.00	8.00	8.00	8.00	8.00

H1: guarded controller without memory
A3: the same base controller with balanced, feasible memory

Mean modeled delay in milliseconds; 1,200 windows per cell

Memory after the same guard



Mixed load: A3 minus H1 = +0.473 ms.
The added memory did not improve this comparison.

Mean modeled delay in ms; bootstrap intervals are in the report

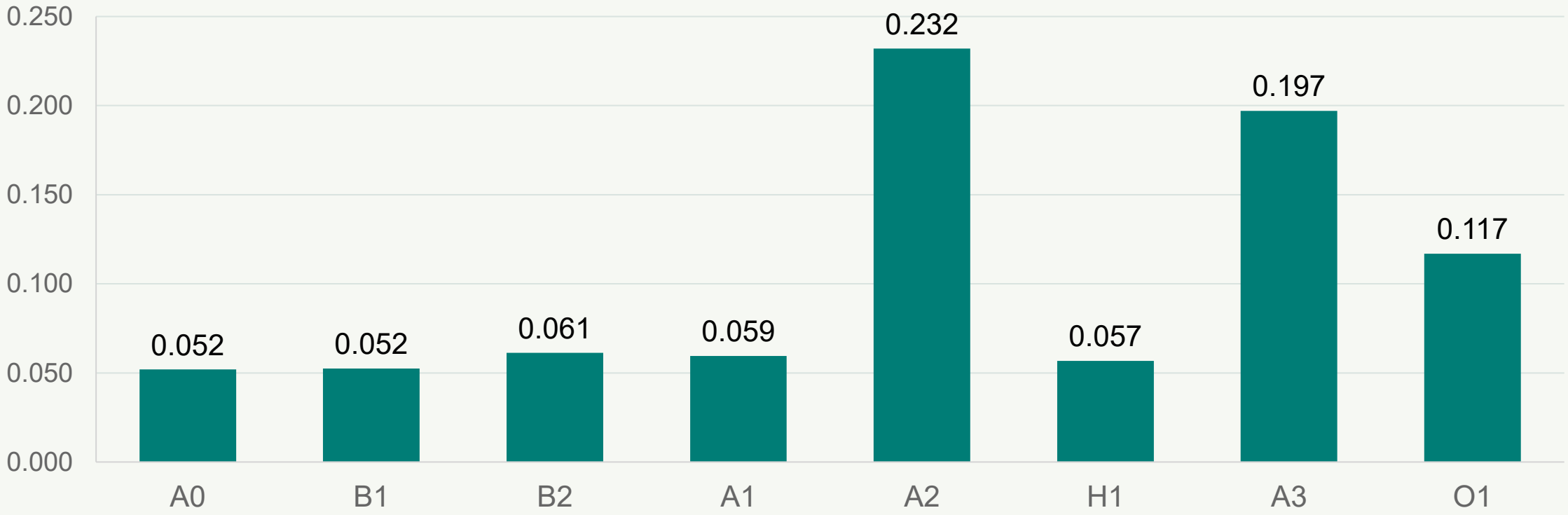
Aggregate performance and feasibility

ID	Delay ms	Goodput Mbps	Mean loss %	Vetoes
A0	24.666	12.929	7.808	0
B1	19.829	13.041	6.941	0
B2	22.180	12.986	7.363	1887
A1	16.565	12.875	7.902	0
A2	16.964	12.824	8.314	134
H1	16.649	12.979	7.291	0
A3	16.829	12.960	7.443	0
O1	16.700	12.987	7.246	0

A2: 134 rejected proposals. A3: 0. Performance remains a separate question.

Equal weights across six scenarios and ten seeds

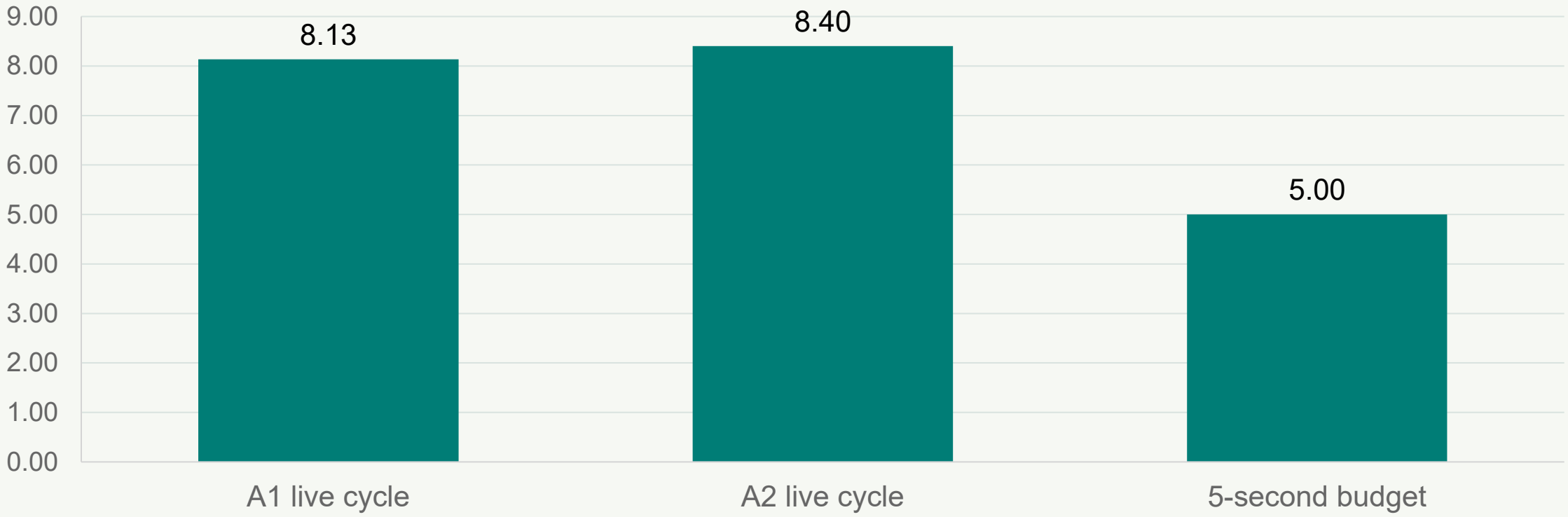
Measured offline computation overhead



Full suite: 11.33 seconds in .venv.
Timing depends on this computer and excludes external API waiting.

Mean control-loop computation in ms; excludes final memory flush

Live API latency and the control deadline



48/48 fresh API calls succeeded. Pacing keeps the demo reliable; this execution does not establish a five-second real-time controller.

Fresh Gemini pilot executed in the isolated .venv

Reproduce and inspect the evidence

```
.\run.ps1 all
.\run.ps1 benchmark
.\run.ps1 dashboard
```

README → project guide → architecture → experimental protocol

CSV + JSONL evidence · Editable DOCX and PPTX · Report and slide PDFs

Run from C:\Users\RAVIPRAKASH\SDN\SDN\selfx